SOLUTIONS: ESTIMATION AND FAULT DETECTION

1. Solution

a) Many ways exist to obtain a transfer function. As a fast way, the transfer function $G_{yr}(s)$ can be obtained as follows (by using the forward gains in the numerator and loops gains in the denumerator):

$$G_{yr}(s) = \frac{G_1(s)G_2(s)G_3(s)G_4(s) + G_1(s)G_4(s)}{1 + KG_1(s)G_2(s)G_3(s)G_4(s) + KG_1(s)G_4(s) - G_3(s)G_4(s)}.$$

[3 Marks]

b) Based on the closed-loop block-scheme of the system, it can be said that

$$\begin{aligned}(R(s) - X_3(s))\frac{1}{s+5} &= X_1(s),\\ (5X_1(s) + X_3(s))\frac{1}{s} &= X_2(s),\\ (X_1(s) + X_2(s))\frac{1}{s+3} &= X_3(s)\end{aligned}$$

yielding

$$\begin{aligned}\dot{x}_1 &= -5x_1 - x_3 + r,\\ \dot{x}_2 &= 5x_1 + x_3,\\ \dot{x}_3 &= x_1 + x_2 - 3x_3.\end{aligned}$$

Therefore, the system state matrix and output matrix are as follows:

$$A = \begin{bmatrix}-5 & 0 & -1\\ 5 & 0 & 1\\ 1 & 1 & -3\end{bmatrix}, B = \begin{bmatrix}1\\0\\0\end{bmatrix}, C = \begin{bmatrix}0 & 0 & 1\end{bmatrix}.$$

[3 Marks]

c) Based on the obtained A and C, the observability can be checked by checking the rank of the observability matrix as follows:

$$\text{rank}\left(\begin{bmatrix}C\\CA\\CA^2\end{bmatrix}\right) = \text{rank}\left(\begin{bmatrix}0 & 0 & 1\\1 & 1 & -3\\-3 & -3 & 9\end{bmatrix}\right) = 2.$$

Moreover, to determine a basis for the nonobservable vector subspace, we should find v such that

$$\begin{bmatrix}C\\CA\\CA^2\end{bmatrix} v = 0_{3\times 1}.$$

Therefore,

$$v = \begin{bmatrix}1\\-1\\0\end{bmatrix}.$$

Answers

[3 Marks]

d) To check the unobservable pole, we should find the pole (an eigenvalue of A) not appearing in the output of the system. First we obtain the eigenvalues of A which are the roots of $\det(sI-A)$ as follows:

$$\det(sI-A) = \det\begin{bmatrix} s+5 & 0 & 1 \\ -5 & s & -1 \\ -1 & -1 & s+3 \end{bmatrix} = s(s+3)(s+5).$$

Therefore, the eigenvalues of A are

$$\text{eig}(A) = \{-5,-3,0\}.$$

Moreover, it can be said that

$$(sI-A)^{-1} = \begin{bmatrix} \frac{s^2+3s-1}{s(s+3)(s+5)} & \frac{-1}{s(s+3)(s+5)} & \frac{-1}{(s+3)(s+5)} \\ \frac{5s+16}{s(s+3)(s+5)} & \frac{s^2+8s+16}{s(s+3)(s+5)} & \frac{1}{(s+3)(s+5)} \\ \frac{1}{s(s+3)} & \frac{1}{s(s+3)} & \frac{1}{s+3} \end{bmatrix}.$$

Now, we check the roots appearing in the output as follows:

$$C(sI-A)^{-1} = \begin{bmatrix} \frac{1}{s(s+3)} & \frac{1}{s(s+3)} & \frac{1}{s+3} \end{bmatrix}.$$

Therefore, we can conclude that $s=-5$ is the unobservable pole of the closed-loop system.

[4 Marks]

e) We should determine a basis T for the orthogonal complement of $\ker(Q)$ where

$$Q = \begin{bmatrix} C \\ CA \\ CA^2 \end{bmatrix}.$$

Considering the nonobservable vector subspace of Q, the matrix T can be chosen as

$$T = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}.$$

Now, we consider a similarity transformation matrix T and a new variable z such that $x = Tz$. Therefore, by defining $z = \begin{bmatrix} z_1 & z_2 & z_3 \end{bmatrix}^\top$,

$$\begin{bmatrix} \dot{z}_1 \\ \dot{z}_2 \\ \dot{z}_3 \end{bmatrix} = T^{-1}ATz + T^{-1}Br,$$

$$y = CTz.$$

It can be said that

$$T^{-1} = \begin{bmatrix} 0.5 & -0.5 & 0 \\ 0 & 0 & 1 \\ 0.5 & 0.5 & 0 \end{bmatrix}.$$

Answers

Thus,

$$\begin{bmatrix} \dot{z}_1 \\ \dot{z}_2 \\ \dot{z}_3 \end{bmatrix} = \begin{bmatrix} -5 & -1 & -5 \\ 0 & -3 & 2 \\ 0 & 0 & 0 \end{bmatrix} z + \begin{bmatrix} 0.5 \\ 0 \\ 0.5 \end{bmatrix} r,$$
$$y = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix}$$

in which the subsystem described by $\begin{bmatrix} \dot{z}_2 & \dot{z}_3 \end{bmatrix}^\top$ is observable.

[4 Marks]

f) Output feedback cannot change the observability of the system for any K.

[3 Marks]

Answers

2. Solution

a) Since $R = 1$ and $L = 1$, by considering $x_1 = V_c$ and $x_2 = I$, it can be said that

$$\begin{aligned} \dot{x}_1 &= x_2, \\ \dot{x}_2 &= -x_2 - x_1 + u, \\ y &= x_1. \end{aligned}$$

[3 Marks]

b) By defining

$$A = \begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix}, B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, C = \begin{bmatrix} 1 & 0 \end{bmatrix},$$

the full-order state observer can be designed as follows:

$$\dot{\hat{x}} = A\hat{x} + Bu + L(y - C\hat{x}).$$

The dynamics of e can be stated as follows:

$$\dot{e} = Ae - LCe$$

implying

$$F = A - LC.$$

By defining $L = \begin{bmatrix} l_1 \\ l_2 \end{bmatrix}$, F is as follows:

$$F = \begin{bmatrix} -l_1 & 1 \\ -1 - l_2 & -1 \end{bmatrix}.$$

Therefore, it is easy to say that if $l_1 = 1$ and $l_2 = -1$, the eigenvalues of F are as -1 and -1.

[4 Marks]

c) From the Laplace transformation of $\dot{e} = Fe$, it can be said that

$$(sI - F)E(s) = e(0).$$

Thus,

$$e(t) = \mathscr{L}^{-1}\{(sI - F)^{-1}\}e(0)$$

which based on F, can be obtained as

$$e(t) = \mathscr{L}^{-1}\left\{ \begin{bmatrix} s+1 & -1 \\ 0 & s+1 \end{bmatrix}^{-1} \right\} e(0).$$

Therefore,

$$e(t) = \mathscr{L}^{-1}\left\{ \begin{bmatrix} \frac{1}{s+1} & \frac{1}{(s+1)^2} \\ 0 & \frac{1}{s+1} \end{bmatrix} \right\} e(0)$$

and

$$e(t) = \begin{bmatrix} e^{-t} & te^{-t} \\ 0 & e^{-t} \end{bmatrix} e(0).$$

As a result

$$\varepsilon(t) = Ce(t) = \begin{bmatrix} e^{-t} & te^{-t} \end{bmatrix} e(0).$$

[5 Marks]

Answers

d) By considering answer to Question 2-b), in the presence of faults in $u(t)$, we have

$$\dot{e} = Fe + Bf.$$

It can be said that

$$(sI - F)E(s) = e(T_0) + B\frac{f}{s}.$$

Thus,

$$e(t) = \mathscr{L}^{-1}\{(sI-F)^{-1}\}e(T_0) + \mathscr{L}^{-1}\left\{(sI-F)^{-1}B\frac{f}{s}\right\}$$

which based on F, can be obtained as

$$e(t) = \mathscr{L}^{-1}\left\{\begin{bmatrix} s+1 & -1 \\ 0 & s+1 \end{bmatrix}^{-1}\right\}e(T_0) + \mathscr{L}^{-1}\left\{\begin{bmatrix} s+1 & -1 \\ 0 & s+1 \end{bmatrix}^{-1} B\frac{f}{s}\right\}.$$

Therefore,

$$e(t) = \mathscr{L}^{-1}\left\{\begin{bmatrix} \frac{1}{s+1} & \frac{1}{(s+1)^2} \\ 0 & \frac{1}{s+1} \end{bmatrix}\right\}e(T_0) + \mathscr{L}^{-1}\left\{\begin{bmatrix} \frac{1}{s(s+1)} & \frac{1}{s(s+1)^2} \\ 0 & \frac{1}{s(s+1)} \end{bmatrix}\right\}Bf$$

and

$$e(t) = \begin{bmatrix} e^{-t} & te^{-t} \\ 0 & e^{-t} \end{bmatrix}e(0) + \begin{bmatrix} 1-e^{-t} & 1-e^{-t}-te^{-t} \\ 0 & 1-e^{-t} \end{bmatrix}Bf.$$

As a result,

$$\varepsilon(t) = Ce(t) = \begin{bmatrix} e^{-t} & te^{-t} \end{bmatrix}e(0) + (1-e^{-t}-te^{-t})f.$$

In this condition, if $|\varepsilon(t)| > \tilde{\varepsilon}(t)$, a fault is detected where

$$\tilde{\varepsilon}(t) = 5e^{-t}\sqrt{1+t^2}.$$

[4 Marks]

e) We should define the fault signal as a new state $x_3 = f$ such that since $u_i = V_i + f$, the system state space equation can be restated as follows:

$$\begin{aligned} \dot{x}_1 &= x_2, \\ \dot{x}_2 &= -x_2 - x_1 + V_i + x_3, \\ \dot{x}_3 &= 0, \\ y &= x_1. \end{aligned}$$

Therefore, the modified state matrix, input matrix, and output matrix can be defined as

$$\bar{A} = \begin{bmatrix} 0 & 1 & 0 \\ -1 & -1 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \bar{B} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \bar{C} = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}. \tag{2.1}$$

Now, the observability can be checked by checking the rank of the observability matrix as follows:

$$\text{rank}\left(\begin{bmatrix} \bar{C} \\ \bar{C}\bar{A} \\ \bar{C}\bar{A}^2 \end{bmatrix}\right) = \text{rank}\left(\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & -1 & 1 \end{bmatrix}\right) = 3.$$

Therefore, the fault is observable. Now, it is sufficient to design an observer for the system (2.1) to estimate the state vector $\hat{x} = \begin{bmatrix} \hat{x}_1 & \hat{x}_2 & \hat{x}_3 \end{bmatrix}^\top$.

[4 Marks]

Answers

3. Solution

a) For $a \neq 0$, since $H \neq 0$, according to

$$P(t+1) = F\Big[P(t) - P(t)H^\top \big(V_2 + HP(t)H^\top\big)^{-1} HP(t)\Big]F^\top + V_1,$$

as $t \to \infty$, $P(t)$ converges to a constant quantity.

[4 Marks]

b) The general algebraic Riccati equation is

$$P(t+1) = F\Big[P(t) - P(t)H^\top \big(V_2 + HP(t)H^\top\big)^{-1} HP(t)\Big]F^\top + V_1, t = 1, 2, \ldots.$$

- For $F = \frac{1}{2}$, $H = \frac{1}{2}$, $V_1 = 1$, and $V_2 = 1$, we can say that the solution $\bar{P}_1$ for $P(t)$ satisfies

$$\bar{P}_1 = \frac{1}{4}\Big[\bar{P}_1 - \frac{1}{4}\big(1 + \frac{1}{4}\bar{P}_1\big)^{-1}\bar{P}_1^2\Big] + 1$$

which can be simplified as follows:

$$16\bar{P}_1 = 4\bar{P}_1 - \big(1 + \frac{1}{4}\bar{P}_1\big)^{-1}\bar{P}_1^2 + 16$$

which can be simplified as

$$\bar{P}_1^2 + 2\bar{P}_1 - 4 = 0.$$

Therefore, the solutions of $\bar{P}_1$ are

$$\bar{P}_{11} = -3.236, \bar{P}_{12} = 1.236$$

which only the positive solution is acceptable.

- For $F = -\frac{1}{3}$, $H = \frac{1}{2}$, $V_1 = 1$, and $V_2 = 1$, we can say that the solution $\bar{P}_2$ for $P(t)$ satisfies

$$\bar{P}_2 = \frac{1}{9}\Big[\bar{P}_2 - \frac{1}{4}\big(1 + \frac{1}{4}\bar{P}_2\big)^{-1}\bar{P}_2^2\Big] + 1$$

which can be simplified as follows:

$$36\bar{P}_2 = 4\bar{P}_2 - \big(1 + \frac{1}{4}\bar{P}_2\big)^{-1}\bar{P}_2^2 + 36$$

which can be simplified as

$$9\bar{P}_2^2 + 23\bar{P}_2 - 36 = 0.$$

Therefore, the solutions of $\bar{P}_2$ are

$$\bar{P}_{21} = -3.651, \bar{P}_{22} = 1.096$$

which only the positive solution is acceptable.

[4 Marks]

Answers

c) According to the obtained $\bar{P}$

$$\bar{K} = F\bar{P}H^{\top}(V_2 + H\bar{P}H^{\top})^{-1}$$

which for $\bar{P}_1 = 1.236$, $F = \frac{1}{2}$, $H = \frac{1}{2}$, and $V_2 = 1$,

$$\bar{K}_1 = 0.2361$$

and for $\bar{P}_2 = 1.096$, $F = -\frac{1}{3}$, $H = \frac{1}{2}$, and $V_2 = 1$,

$$\bar{K}_2 = -0.1434.$$

Therefore, the difference equation yielding the one-step ahead optimal steady-state Kalman prediction is as follows:

$$\hat{x}(t+1|t) = \frac{1}{2}\hat{x}(t|t-1) + 0.2361\big(y(t) - \frac{1}{2}\hat{x}(t|t-1)\big).$$

and

$$\hat{x}(t+1|t) = -\frac{1}{3}\hat{x}(t|t-1) - 0.1434\big(y(t) - \frac{1}{2}\hat{x}(t|t-1)\big).$$

[3 Marks]

d) According the system model and the Kalman filter difference equation, the dynamical system is sketched in Fig. 3.1

[3 Marks]

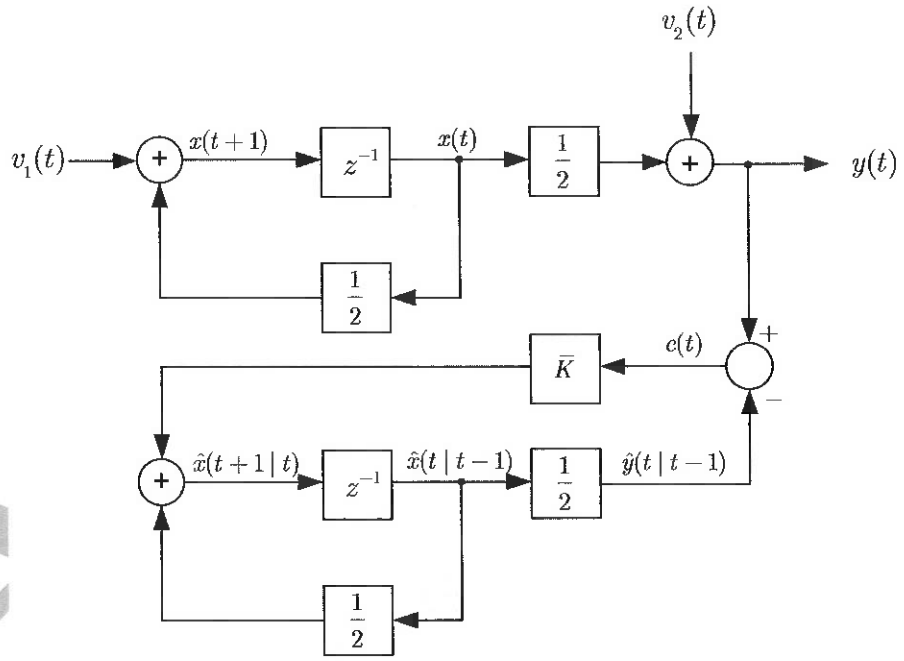


Figure 3.1 The system configuration in Question 3.

e) The transfer functions from the noises $v_2(t)$ to the estimation error $e(t) = y(t) - \frac{1}{2}\hat{x}(t|t-1)$ are as follows:

$$G_{v_2 e}^{(2)} = \frac{Y(z)}{V_2(z)} \cdot \frac{E(z)}{Y(z)} = 1 \cdot \frac{1}{1 + \frac{K}{2z-2a}}.$$

or

$$G_{v_{2e}}^{(2)} = \frac{Y(z)}{V_2(z)} \cdot \frac{E(z)}{Y(z)} = \frac{2z-2a}{2z-2a+K}.$$

Hence, denoting by $\Gamma_2(\omega)$ the spectrum of $v_2(t)$ and by $\Gamma_e(\omega)$ the spectrum of $e(t)$, one gets:

$$\Gamma_e(\omega) = \|G_{v_{2e}}^{(2)}(e^{j\omega})\|^2 \Gamma_2(\omega)$$

from which a spectral analysis can be carried out to ascertain the spectral characteristics of $e(t)$ as a function of the value of a.

[6 Marks]

Answers

4. Solution

a) Based on the system state-space equation, we have

$$A=\begin{bmatrix}-1 & 1 & 0\\ 0 & 0 & 1\\ -2 & -1 & 0\end{bmatrix}, C=\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 1\end{bmatrix}, B=\begin{bmatrix}0\\ 0\\ 1\end{bmatrix}.$$

It can be said that

$$\det(sI-A)=s^3+s^2+s+3.$$

Therefore,

$$(sI-A)^{-1}=\frac{1}{s^3+s^2+s+3}\begin{bmatrix}s^2+1 & s & 1\\ -2 & s(s+1) & s+1\\ -2s & -(s+3) & s(s+1)\end{bmatrix}$$

and

$$C(sI-A)^{-1}=\frac{1}{s^3+s^2+s+3}\begin{bmatrix}s^2+1 & s & 1\\ -2s-2 & s^2-3 & s^2+2s+1\end{bmatrix}.$$

Moreover, the roots of s^3+s^2+s+3 are $\{-1.5747, 0.2874\pm 1.35i\}$. Therefore, all the poles appear in the output, and the system is fully observable.

Moreover,

$$G_{yu}(s)=C(sI-A)^{-1}B=\frac{1}{s^3+s^2+s+3}\begin{bmatrix}1\\ s^2+2s+1\end{bmatrix}.$$

[5 Marks]

b) Since the rank of C is 2, the order of the reduced order state observer is $n-p=3-2=1$.

[3 Marks]

c) It is not required to estimate x_1 as it is measurable. If the reduced-order state observer estimate x_2, x_3 can be estimated via the following equation:

$$\hat{x}_3=y_2-\hat{x}_2.$$

By defining $x=\begin{bmatrix}x_1 & x_2 & x_3\end{bmatrix}^\top$, let us consider the following state transformation:

$$z=\begin{bmatrix}z_1\\ z_2\end{bmatrix}=Tx=\begin{bmatrix}I_2 & \begin{bmatrix}0\\ 1\end{bmatrix}\\ 0_{1\times 2} & 1\end{bmatrix}x.$$

Therefore, we have

$$\dot{z}=\tilde{A}z+\tilde{B}u,$$
$$y=\tilde{C}z$$

where

$$\tilde{A}=TAT^{-1}=\begin{bmatrix}-1 & 1 & -1\\ -2 & -1 & 2\\ -2 & -1 & 1\end{bmatrix}, \tilde{B}=TB=\begin{bmatrix}0\\ 1\\ 1\end{bmatrix}, \tilde{C}=CT^{-1}=\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\end{bmatrix}.$$

Answers

Therefore, the reduced-order state observer has the following form:

$$\dot{\xi} = F\xi + Hy + Gu,$$
$$\hat{z}_2 = \xi + Ly$$

where $F = -1$ and L, G, and H can be obtained as follows:

$$\begin{aligned} L\tilde{A}_{12} - \tilde{A}_{22} &= -F, \\ L\tilde{A}_{11} + H - \tilde{A}_{21} &= FL, \\ G + L\tilde{B}_1 - \tilde{B}_2 &= 0. \end{aligned} \tag{4.1}$$

According to the definition of z and $\tilde{A}$,

$$\tilde{A}_{11} = \begin{bmatrix} -1 & 1 \\ -2 & -1 \end{bmatrix}, \tilde{A}_{12} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}, \tilde{A}_{21} = \begin{bmatrix} -2 & -1 \end{bmatrix}, \tilde{A}_{22} = \begin{bmatrix} 1 \end{bmatrix}, \tilde{B}_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \tilde{B}_2 = 1.$$

Therefore, from (4.1), it follows that

$$\begin{bmatrix} l_1 & l_2 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix} - 1 = 1,$$
$$\begin{bmatrix} l_1 & l_2 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ -2 & -1 \end{bmatrix} + \begin{bmatrix} h_1 & h_2 \end{bmatrix} - \begin{bmatrix} -2 & -1 \end{bmatrix} = -\begin{bmatrix} l_1 & l_2 \end{bmatrix},$$
$$G + \begin{bmatrix} l_1 & l_2 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} - 1 = 0$$

which has mutiple solution.

[8 Marks]

d) Based on the system model and the observer, the block-scheme of the system and of the observer is depicted in Fig. 4.1.

[4 Marks]

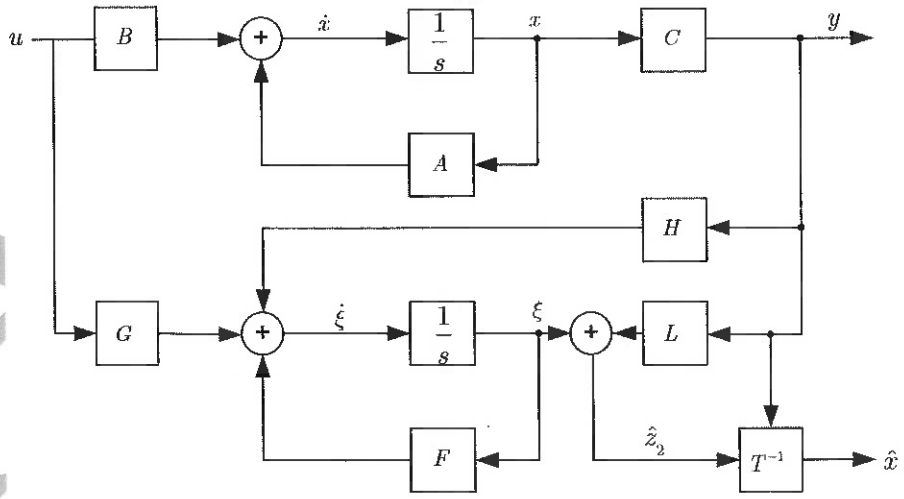


Figure 4.1 The system/observer block-scheme in Question 4-d.

ANSWERS